

ORGANIZATIONAL BEHAVIOR RESILIENCE MODEL - A COMPARATIVE STUDY

Author(s)*: Raluca Petronela LĂZĂRESCU ¹, Adrian VÎLCU ²
Position: Assist. Prof., PhD ¹, Assistant Prof., PhD ²
University: "Gheorghe Asachi" Technical University of Iași
Address: Iași, Str. Prof. Dr. Doc. Dimitrie Mangeron, Nr. 29, România
Email: raluca-petronela.lazarescu@academic.tuiasi.ro¹, adrian.vilcu@academic.tuiasi.ro²,
Webpage: <http://www.tuiasi.ro/>

Abstract

Purpose – the objective of the study was to identify the mechanism through which young adults faced the pandemic period in a way to maintain a subjective well-being and to perceive stress. The paper presents a compared study following the adaptative reactions of individuals that have to change during the pandemic period, in the direction of a higher resilience level.

Methodology/approach - the authors used the CD-RISC which is the suggested self-report scale for measuring trait resiliency. The modeling tool used is the Principal Component Analysis (PCA) method to reduce the terms of resilience influence in the important way. The authors compare two models for the two groups of respondents.

Findings – conducting a compared study between pre and post pandemic period regarding the resilience; Identification of the differences may occur in order to face the traumatic experiences at the organizational behavior level.

Research limitations/implications – the models are limited to the type of respondents approached in the research, but the methodology has a practical character in the research of social phenomena.

Practical implications – the paper represents a quantitative analysis tool of some psycho-social phenomena characteristic of young adults.

Originality/value - the research work is a useful quantitative analysis tool in understanding the evolution of the phenomenon of resilience in the case of young adults and the factors that determine it.

Key words: Principal Components Analysis, resilience factor, statistical methods

Introduction

According to Rutter (1985), resilience is the ability to recover or adaptive coping despite adversity or exposure to stress factors.

There is a consensus that resilience plays a vital role in the way a person experiences traumatic events (Arnau, 2002; Carle & Chassin, 2004; Juffer, Stams, & van IJzendoorn, 2004), and that involves adapting it successfully despite the situational risk factors (Carle & Chassin).

The interest in resilience was initially dependent on negative external changes. Carpenter, Walker, Anderies and Abel (2001) identify resilience in the context of a system's optimal functioning, which involves significant positive changes that lead to the development of the individual.

More and more researchers are adopting a definition of resilience, often intuitive, describing resilience as the ability to recover from a shock (Walker, Carpenter, and Kinzig, 2004). Carpenter & Brock (2008) insists on understanding the process of reorganizing a system with the aim of optimal functioning, following changes from inside or outside.

Resilience may represent a mechanism that explains how organizational behaviour functions under stress conditions, such as the coronavirus pandemic.

The study's objective was to identify the mechanism through which young adults faced the pandemic period to maintain subjective well-being and perceive stress.

The paper presents a comparison study following the adaptative reactions of individuals that have to change during the pandemic period in the direction of a higher resilience level.

The authors invited two student groups enrolled in courses at Gheorghe Asachi Technical University of Iași, Romania, Economical Engineering specialization (n1=137 and n2=123) to participate in an online questionnaire. Students who expressed interest were provided with a link to complete the questionnaire. The research using Google Forms was administered in February – March 2016 before the COVID pandemic and May June 2022 during the post-COVID-19 pandemic. Informed consent was obtained virtually on the first page of the online questionnaire. Participation in the study was voluntary, and participants did not receive compensation.

To measure the level of trait resiliency in participants, the authors used the CD-RISC, the suggested self-report scale for measuring trait resiliency (Connor & Davidson, 2003).

The modelling tool used is the PCA (Principal Component Analysis) method to reduce the terms of resilience influence in a significant way. The authors compare two models for the two groups of respondents with statistical methods.

Organisation of research

The research is structured on the steps of comparative analysis of the data obtained from the completion of some questionnaires by a sample of respondents.

The research methodology includes the following aspects:

- Defining the research problem and motivating the theme. The problem is the comparative analysis of the resilience factor in the organisational framework before and after the pandemic period. The topic is actual and necessary for understanding the dynamics of the resilience factor after two years of an intense health crisis. A second reason for the importance of the studied theme is that a characteristic model, well-made on the specified type of respondents, leads to a better understanding of the evolution of the state of resilience, and a prediction of its dynamics can be made. We know that the effects of two years of health crisis are not clearly seen in such a short time. Still, these models can be compared with those generated in the near or more distant future, developing a clearer understanding of the phenomenon.
- I was choosing the questionnaire. For this analysis, a Cd-RISC professional questionnaire was selected and used in many international studies for the analysis of the resilience factor in different response groups. The questionnaire is applied to two groups of respondents, in the same social category and in the same age range, the application interval being six years, 2016 - 2022. The analysis was focused on two different groups of young adult respondents, students at the Technical University "Gh. Asachi" from Iasi, years 1 and 2.
- Implementation of input data (input variables) of the model and output variables. The 25 items form the input variables. They are grouped into aspects that will be analysed from the point of view of the relationship with the resilience factor. The output variable, the resilience factor, was defined by summing the values of the input variables (Taherdoost, 2020).
- Descriptive statistical analysis on the output variable related to the two groups of respondents. The distribution of the resilience factor in the two groups of respondents is analysed, determining the comparative values of the distribution parameters. The distributions of the items are analysed to choose the types of statistical tests for the analysis.
- Comparative statistical analysis on the most important aspects that characterise the resilience factor. The non-parametric Mann-Whitney U test was applied to analyse the frequency histograms of the ranks for the essential elements of resilience.
- Implementation of factorial models for each group of respondents to observe the most important items within the model - PCA method (Sidharth et al. I, 2017).

Experimental results

1. Descriptive analysis of resilience factors for the two groups

Descriptive analysis for the 2016 group.

The resilience variable is made up of the sum of all 25 items. The histogram of the series is presented in figure 1.

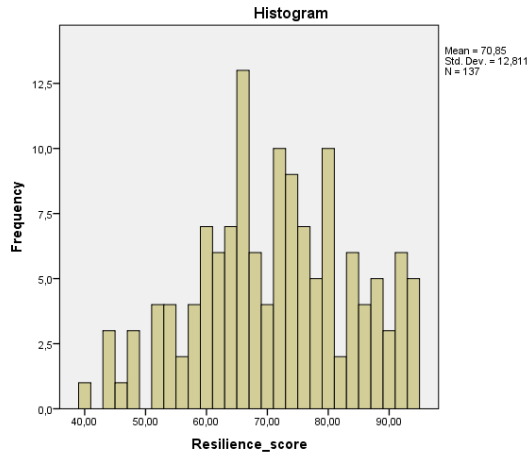


Fig.1 Histogram of the resilience variable for the 2016 group

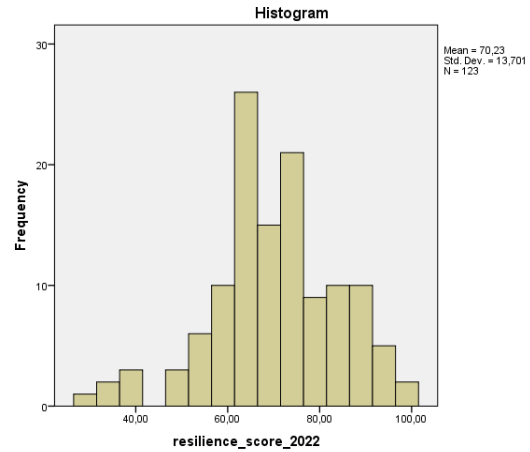


Fig.2 Histogram of the resilience variable for the 2022 group

The K-S test to analyse the distribution of the resilience variable (table 1) specifies that the variable is normally distributed.

Table 1. Resilience variable for 2016 group – normality test

K-S test	Statistic	df	Sig.
Resilience_score	,049	137	,200*

The descriptive statistics data are presented in table 2.

Table 2. Resilience variable for 2016 group – descriptive statistic

		Statistic	Std. Error
Resilience_score	Mean	70,8540	1,09453
	Median	71,0000	
	Variance	164,126	
	Std. Deviation	12,81115	
	Minimum	40,00	
	Maximum	94,00	
	Range	54,00	
	Interquartile Range	18,00	
	Skewness	-,125	,207
	Kurtosis	-,590	,411

The value of the average parameter $m=70.85$ is representative of the entire series. This value represents a high level of termination (value in the range [70;100]), which is not surprising.

Descriptive analysis for the 2022 group. For this respondent group, the resilience variable maintains the normality of the distribution (table 3).

Table 3. Resilience variable for 2022 group – normality test

K-S test	Statistic	df	Sig.
Resilience_score	,073	123	,165

The histogram of the resilience variable for the 2022 group is presented in figure 2.

The descriptive statistics data are presented in table 4.

Table 4. Resilience variable for 2022 group – descriptive statistic

		Statistic	Std. Error
Resilience score	Mean	70,22	1,23
	Median	70,75	
	Variance	187,71	
	Std. Deviation	13,70	
	Minimum	29,00	
	Maximum	98,00	
	Range	69,00	
	Interquartile Range	16,00	
	Skewness	-,408	,218
	Kurtosis	,520	,433

The mean of the resilient series, representative of the normally distributed series, has a value of 70,2276, a value corresponding to a high level of resilience for the 2022 group.

Comparing the two average values for the resilience factor corresponding to the two groups of respondents, we notice that they are approximately equal. The only observed difference is a greater degree of variability for the 2022 group. The resulting conclusions can be explained by the nature of the two groups of respondents.

2. Analysis of the frequency ranks of the resilience variable

If the level of resilience for the two groups is the same, we will analyse their distribution frequencies. Since the resilience variable consists of non-normally distributed series, we will use the Mann-Whitney U test to analyse the ranks of the series values (Yonghwan, 2018). The resilience variable for the two groups of respondents (before and after the pandemic) has the same distribution (figure 3).

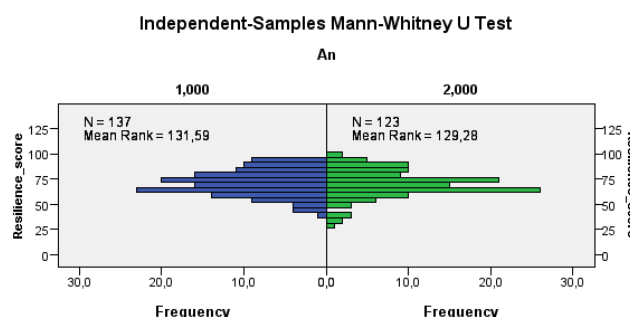


Fig. 3 Distribution of the Resilience variable for the two groups of respondents

According to the non-parametric Mann-Whitney U test (Qing He, 2020), the null hypothesis H0 is accepted: there is no difference between the distributions of the value ranks of the two analysed groups, with a confidence level of 95% (p-value=0,804). Furthermore, it can be seen that the average of group 1 has a slightly higher rank level than the average rank of group 2 (131,59 vs 129,28), which strengthens the conclusion that the level of resilience has not changed for the questionnaire respondents (figure 3).

This conclusion is consistent with natural logic, namely that the respondents are young adults who went through the pandemic protected by their families.

The only aspect that is different is item rz24, "I act in such a way as to achieve my goals, no matter what obstacles I encounter in the way", which has different distributions (figure 4). There is a difference in the positioning of the average value of 120 vs 142 in favour of group 2, which went through the pandemic period.

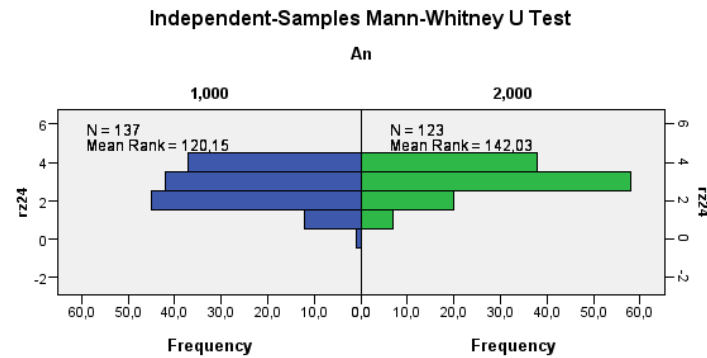


Fig. 4 Distribution of the rz24 aspect for the two groups of respondents

Three aspects considered to be essential and defining for the resilience score for the two groups of respondents were analysed: "acceptance of change" (table 5) (figure 5), "self-control" (table 6) (figure 6) and "tenacity" (table 7).

Table 5. „Acceptance of change” aspect, with its items

Aspect	Item	Description
Acceptance of change	1	I can adapt when changes occur.
	4	I can handle anything that comes my way.
	5	Past successes give me confidence in front of new challenges and difficulties.
	8	I tend to recover quickly from illness, injury or any other hardships.

For the acceptance of change aspect, it can be observed that the structure of the resilience level is approximately the same (figure 5). The ranks of the two mean values are roughly equal. The statistical analysis does not surprise us.

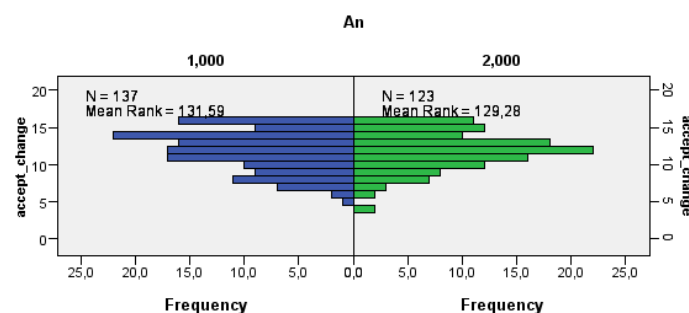


Fig. 5 Distribution of resilience ranks for the "Acceptance of change" aspect

Table 6. „Acceptance of change” aspect, with its items

Aspect	Item	Description
Self-control	21	I have a clear purpose in life.
	22	I feel in control of my life.

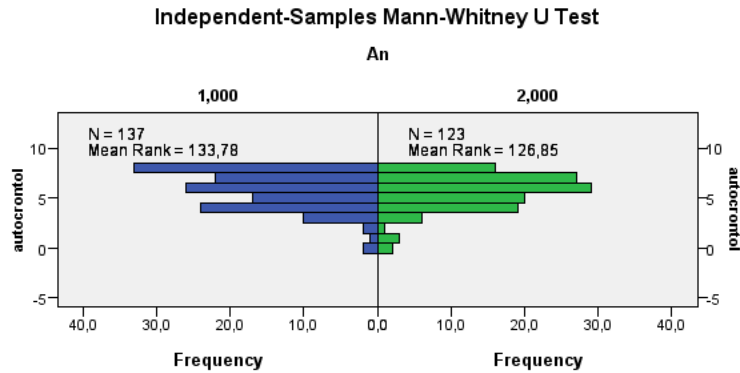


Fig. 6 Distribution of resilience ranks for the "Self-control" aspect

Table 7. Tenacity aspect, with its items

Aspect	Item	Description
Tenacity	10	I try my best, no matter what the result is.
	12	Even when things seem hopeless, I don't give up.
	16	I am not easily discouraged in case of failure.
	24	I act in such a way as to achieve my goals, no matter what obstacles I encounter on the way.

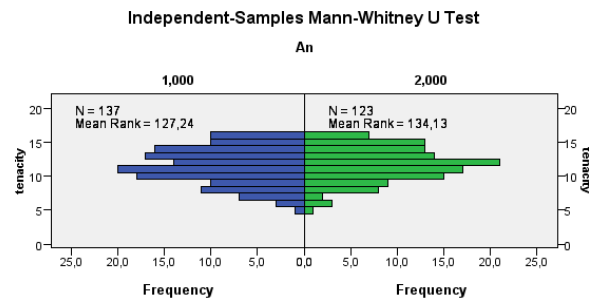


Fig. 7 Distribution of resilience ranks for the "Tenacity" aspect

There is a difference in the rank for the average indicator of 127 for the first group of respondents, lower than the value of 134, which represents the average rank of group 2, which leads to the conclusion that after the pandemic period, the level of tenacity increased. The respondents are more persistent, and we can consider, by correspondence with the tenacity of the materials, that the pandemic period generated an increase in the level of resilience from the point of view of tenacity.

3. Analysis of the resilience score component

For this analysis, we made a factorial reduction to the most important aspects that make up resilience (Zhang, 2015). The method used is CPA and is applied separately to the two groups of respondents (Laxmi, 2016).

After applying the CPA method reducing the 25 items to 7 factors, for 2016 group (table 8) and 2022 group (table 9) and determining the most important items that make up these factors, we obtained the following results (table 10).

Table 8. Application of the CPA method to the data of the 2016 group

Component	Initial Eigenvalues			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	7,152	28,609	28,609	3,915	15,659	15,659
2	2,459	9,835	38,444	3,674	14,695	30,354
3	1,476	5,904	44,348	2,310	9,238	39,592
4	1,349	5,397	49,744	1,918	7,673	47,265
5	1,275	5,101	54,845	1,452	5,808	53,074
6	1,142	4,568	59,413	1,380	5,518	58,592
7	1,053	4,211	63,624	1,258	5,033	63,624

Table 9. Application of the CPA method to the data of the 2022 group

Component	Initial Eigenvalues			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	8,510	34,042	34,042	3,960	15,840	15,840
2	1,852	7,409	41,450	3,631	14,525	30,364
3	1,462	5,848	47,299	2,415	9,661	40,026
4	1,200	4,801	52,099	1,746	6,986	47,012
5	1,156	4,624	56,723	1,639	6,554	53,566
6	1,089	4,354	61,078	1,607	6,428	59,994
7	1,041	4,162	65,240	1,312	5,246	65,240

Table 8. The most important items for resilience

Group 2016	Group 2022
rz21	rz16
rz22	rz01
rz25	rz19
rz11	rz22
rz12	rz08
rz17	rz17

The differences are:

In 2016, the most important aspects that determine the level of resilience (in descending order) were: self-control (rz21, rz22), high standards (rz25, rz17), and personal competence (rz11), tenacity (rz12).

The respondents from the year 2022 consider the following aspects to be the most important: tenacity (rz16), acceptance of the change (rz1, rz8), tolerance to negative effects (rz19), self-control (rz22) and high standards (rz17).

The pandemic period changed the order of importance of aspects related to resilience by introducing new elements: "acceptance of change" and "tolerance to negative effects".

Discussions and conclusions

The first conclusion drawn from the research conducted is that the level of the resilience factor, analysed in young adults (19-22 years old), has not changed at all after a difficult period of the health crisis.

Statistical models generated on the two response groups only indicated an increase in the diversity of the important aspects that build the personality factor of the post-pandemic group compared to the pre-pandemic one. There were two new aspects considered by the response of the second group - "acceptance of change" and "tolerance to negative effects". These two aspects are intrinsic, and I think they are not aware of the respondents.

Another aspect that has a different distribution in the formation of the resilience factor is "tenacity", an average of 134 of group 2 compared to 127 (group 1), but we believe that this is not due to a single factor - the pandemic period.

It is very good that the study demonstrated a high level of resilience in the type of respondents analysed, a conclusion that does not violate natural logic - the explanation is that the students come from a protective family environment, without responsibilities.

References

Rutter, M.
1985

Resilience in the face of adversity: Protective factors and resistance to psychiatric disorder. *The British Journal of Psychiatry*, 147, 598–611. <https://doi.org/10.1192/bjp.147.6.598>

Arnau, Randolph C.
2002

Hope: Its measurement and relationships with personality and mental health, Texas A&M University ProQuest Dissertations Publishing, 3060761.

Carle, A., Chassin, L.
2004

Resilience in a community sample of children of alcoholics: Its prevalence and relation to internalizing symptomatology and positive affect, *Journal of Applied Developmental Psychology* 25(5):577-595, DOI:10.1016/j.appdev.2004.08.005

Juffer, F., Stams, G.-J. J. M., & van IJzendoorn, M. H.
2004

Adopted children's problem behavior is significantly related to their ego resiliency, ego control, and sociometric status. *Journal of Child Psychology and Psychiatry*, 45(4), 697–706. <https://doi.org/10.1111/j.1469-7610.2004.00264.x>

Carpenter, S., Walker, B., Anderies, M., J., Abel . N.
2001

From Metaphor to Measurement: Resilience of What to What?, SpringerLink

Walker, B., C. S. Holling, S. R. Carpenter, and A. Kinzig
2004

Resilience, adaptability and transformability in social–ecological systems. *Ecology and Society* 9(2): 5. [online] URL: <http://www.ecologyandsociety.org/vol9/iss2/art5/>

Carpenter, S. R., and W. A. Brock
2008

Adaptive capacity and traps. *Ecology and Society* 13(2): 40. [online] URL: <http://www.ecologyandsociety.org/vol13/iss2/art40/>

Sidharth Prasad Mishra, Uttam Sarkar, Subhash Taraphder
2017

Multivariate Statistical Data Analysis- Principal Component Analysis (PCA), *International Journal of Livestock Research* eISSN : 2277-1964 NAAS Score 5.36, Vol 7 (5) May'17, DOI 10.5455/ijlr.20170415115235

Yonghwan Um
2018

Combining Independent Permutation p Values Associated with Mann-Whitney Test Data, *Journal of The Korea Society of Computer and Information* Vol.23 No.7, pp. 99-104, July 18

Laxmi Shaw

2016

Statistical Features Extraction for Multivariate Pattern Analysis in Meditation EEG using PCA, 978-1-5090-0935-0/16, 2016 IEEE

Qing He, Junyi Liu, Sizhu Wang & Jishuang Yu

2020

The impact of COVID-19 on stock markets, ECONOMIC AND POLITICAL STUDIES 2020, VOL. 8, NO. 3, 275–288 <https://doi.org/10.1080/20954816.2020.1757570>

Hamed Taherdoost

2020

Validity and Reliability of the Research Instrument; How to Test the Validation of a Questionnaire/Survey in a Research, International Journal of Academic Research in Management (IJARM) Vol. 5, No. 3, 2016, Page: 28-36, ISSN: 2296-1747 © Helvetic Editions LTD, Switzerland

Chao Zhang, Behrend Heeren, Martin Rumpf, and William A.P. Smith

2015

Shell PCA: statistical shape modelling in shell space, Mathematics, 2015 IEEE International Conference on Computer Vision (ICCV), DOI:10.1109/ICCV.2015.195